WORKED SOLUTION

STEP II 1994, Question 5

Pure • Counting Real Roots

STEP II 1994, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q5

The key idea

Counting roots of $f(x) = c$ means asking how many times the horizontal line $y = c$ meets the graph of f, so what matters is the shape of f: where it is smallest, and where it is increasing or decreasing. Every expression here is built from two brackets placed symmetrically about a centre point. Shifting the variable so that the centre becomes the origin makes the odd-power terms cancel (for fourth powers) or the even-power terms cancel (for cubes), and the shape becomes obvious.

Solution

Part (i)

Expand by the binomial theorem. In $(x+1)^4$ and $(x-1)^4$ the terms with odd powers of x have opposite signs, so they cancel in the sum:

$$f(x) = (x-1)^4 + (x+1)^4 = 2(x^4 + 6x^2 + 1) = 2x^4 + 12x^2 + 2.$$

So f is even, and $f(x) = 2 + 2x^4 + 12x^2 \geq 2$, with equality only at $x = 0$. For $x \geq 0$, $f'(x) = 8x^3 + 24x > 0$ when $x > 0$, so f is strictly increasing on $[0, \infty)$, and $f(x) \to \infty$ as $x \to \infty$. Since f is continuous it takes every value in $[2, \infty)$ exactly once on $[0, \infty)$; by symmetry it does the same on $(-\infty, 0]$.

- If $c > 2$: there is exactly one root $x = a > 0$, and then $x = -a$ is a second; the two are distinct and there are no others. So there are **exactly two** real roots.
- If $c = 2$: $f(x) = 2$ forces $2x^4 + 12x^2 = 0$, so $x = 0$ is the **only** root.
- If $c < 2$: $f(x) \geq 2 > c$ for all x, so there are **no** roots.

This is exactly what was required.

(Alternatively, $2x^4 + 12x^2 + 2 - c = 0$ is a quadratic in $u = x^2$ with roots $u = -3 \pm \sqrt{8 + c/2}$; only the $+$ root can be non-negative, and it is positive, zero or negative according as $c > 2$, $c = 2$ or $c < 2$.)

Part (ii)

The brackets $x-3$ and $x-1$ are centred on $x=2$, so put $y=x-2$. Then

$$(x-3)^4+(x-1)^4=(y-1)^4+(y+1)^4,$$

which is the expression from part (i) in the variable y. Since $x \mapsto y = x-2$ is a one-to-one correspondence between real numbers, the equation has the same number of real roots as in part (i):

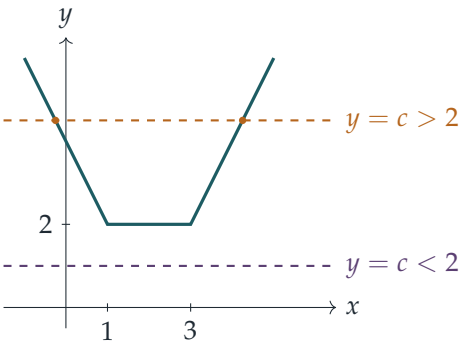
$$\text{two roots if } c>2, \qquad \text{one root } (x=2) \text{ if } c=2, \qquad \text{no roots if } c<2.$$

Part (iii)

Let $g(x)=|x-3|+|x-1|$ and remove the modulus signs on each of the three intervals cut out by $x=1$ and $x=3$:

$$g(x)=\begin{cases}(3-x)+(1-x)=4-2x, & x<1,\\ (3-x)+(x-1)=2, & 1\le x\le 3,\\ (x-3)+(x-1)=2x-4, & x>3.\end{cases}$$

So the graph is a flat-bottomed V: constant at 2 between 1 and 3, strictly decreasing to the left and strictly increasing to the right, with $g>2$ outside $[1,3]$.



- If $c<2$: $g(x)\ge 2>c$ everywhere, so there are **no** roots.
- If $c=2$: every x with $1\le x\le 3$ is a root, and nothing outside, so there are **infinitely many** roots.
- If $c>2$: there is no root in $[1,3]$; on $x<1$, $4-2x=c$ gives $x=2-\frac{c}{2}$, which does lie below 1 since $c>2$; on $x>3$, $2x-4=c$ gives $x=2+\frac{c}{2}>3$. So there are **exactly two** roots, $x=2\pm\frac{c}{2}$.

? **Ask yourself** Why is the answer for $c=2$ so different from part (ii)?

With fourth powers the sum has a single sharp minimum at the centre point, because both terms are strictly convex. With moduli, moving x between 1 and 3 makes one distance grow by exactly as much as the other shrinks, so the sum stays at 2 (the distance between 1 and 3) along the whole interval.

Part (iv)

Again put $y = x - 2$. For cubes it is the even powers of y that cancel:

$$(x-3)^3 + (x-1)^3 = (y-1)^3 + (y+1)^3 = 2y^3 + 6y.$$

Let $h(y) = 2y^3 + 6y$. Then

$$h'(y) = 6y^2 + 6 \geq 6 > 0$$

for every real y, so h is strictly increasing on the whole real line, and hence takes each value at most once. Also $h(y) \to \infty$ as $y \to \infty$ and $h(y) \to -\infty$ as $y \to -\infty$, so, since h is continuous, by the intermediate value theorem it takes every real value at least once.

Hence for **every** real c the equation has **exactly one** real root.

Checks

- At the centre point $x = 2$, both fourth-power expressions equal $1 + 1 = 2$, matching the $c = 2$ cases, and $(2-3)^3 + (2-1)^3 = 0$.
- In part (iv), $c = 8$ should give $x = 3$: indeed $0^3 + 2^3 = 8$, and $h(1) = 2 + 6 = 8$ with $y = 1$.
- In part (iii) with $c = 4$, the roots $x = 0$ and $x = 4$ give $3 + 1 = 4$ and $1 + 3 = 4$.

We confirmed numerically that the expansions hold at random points, and counted sign changes of each function minus c on a fine grid for a range of values of c (for example $c = 1.9$, 2.5, 100 in (i) and (ii), and c from -1000 to 5000 in (iv)): the counts were 0, 2, 2 and always 1, as claimed.

Common slips

- Answering "one root" or "two roots" for $c = 2$ in part (iii): the whole interval $1 \leq x \leq 3$ consists of roots.
- In part (iv), arguing that a cubic "has one or three real roots" without deciding which. The fact that $h'(y) > 0$ everywhere settles it.
- Forgetting to check that $x = 2 - \frac{c}{2}$ really lies in the region $x < 1$ where the formula $4 - 2x$ was used.

Beyond STEP

The arguments here rest on two theorems that a first analysis course proves carefully: the intermediate value theorem (a continuous function on an interval takes every value in between) and the fact that a function with positive derivative is strictly increasing, which follows from the mean value theorem. Part (iii) is also a first taste of convexity and of the ℓ^1 distance: the sum of distances to two points is minimised along the

whole segment between them. Abbott's *Understanding Analysis* develops these ideas rigorously.

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